

Vire Remote Control Buttons and Corresponding Codes

Remote Control Button	Code
Power	18615
Forward key >>	-23271
Backward key <<	31
Button 0	-8161
Button 1	20655
Button 2	-10201
Button 3	-2041
Button 4	12495
Button 5	-20401
Button 6	28815

IR_decoder.ino is used for decoding the keys of the remote control.

Decoded keys are used in ppt_pointer.ino code in the second part, where each key is assigned to a particular function. It is uploaded to Arduino Leonardo for the slide pointer application.

To run the above-mentioned source codes, you need software libraries like Arduino universal IR remote (IRremote.h) and keyboard (Keyboard.h) header files. Download the libraries and install them on the PC at correct locations/folders. You may download header files from GitHub website (<https://github.com/z3t0/Arduino-IRremote>). Do not forget to follow the installation procedure given on github.com

Construction and testing

Connect Arduino to the PC and select the correct COM port in Arduino IDE. Select the correct board from Tools→Board menu in Arduino IDE and upload the programs to Arduino.

For first part, upload IR_decoder.ino code to Arduino. Connect the circuit as shown in Fig. 1. MP3 Vire (21-key) remote was used during testing. Compile and upload the code to Arduino.

Now, point the remote control towards the IR sensor and press any button—the onboard LED of Arduino will blink. Open serial monitor window in Arduino IDE and press any key on the TV/MP3 IR remote.

If everything is alright, when you press a button, you should see a corresponding number on the serial monitor. Each button has a unique corresponding key code, as listed in the table above. Note down the codes and buttons, as these will be used in ppt_pointer.ino code in the second part as explained earlier. (You can use any IR remote and decode the key/button codes using this program.)

Any key of the remote control can be used for pushing the slides forwards or backwards. During testing, >> (forwards) and << (backwards) buttons of the remote were used. Decoded forward key is -23271 and backward key is 31.

After uploading ppt_pointer.ino code to Arduino Leonardo and connecting the sensor, the smart IR pointer will be ready to use.

Open the presentation file with Microsoft PowerPoint or Presentation from Open Office or Libre Office. Open Slide Show page. Every time you press >> key from your MP3 Vire remote, the slide will move forwards and, similarly, on pressing << key, the slide will move backwards. **EFY**



Pamarthi Kanakaraja is assistant professor in ECE department, KL University, Vaddeswaram, Andhra Pradesh

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